

Daily Algebra Drill #6 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.

Section A Rapid Recognition

1. Factorise: $x^3 + 5x^2 + 6x$.

Answer: $x(x+2)(x+3)$

Method: Factor out x : $x(x^2 + 5x + 6) = x(x+2)(x+3)$. Always extract common factors before any other step.

Investigate further: The roots are 0, -2, -3. The product $0 \times (-2) \times (-3) = 0$ confirms $c = 0$ in Vieta. Now: for $x^3 + ax^2 + bx = 0$ (no constant term), $x = 0$ is always a root. What does this mean about the graph of $y = x^3 + 5x^2 + 6x$?

2. Given $x + \frac{1}{x} = \sqrt{5}$, find $x^2 + \frac{1}{x^2}$.

Answer: 3

Method: $(x + \frac{1}{x})^2 = x^2 + 2 + \frac{1}{x^2} = 5$, so $x^2 + \frac{1}{x^2} = 3$.

Investigate further: Note $x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2 = 5 - 2 = 3 < 5$. Continue the chain to find $x^3 + \frac{1}{x^3} = (\sqrt{5})(3) - \sqrt{5} = 2\sqrt{5}$ and $x^4 + \frac{1}{x^4} = 3^2 - 2 = 7$.

3. State the coefficient of xy^2 in $(x+y)^3$.

Answer: 3

Method: $\binom{3}{1} = 3$ (choosing 1 factor to contribute x , the rest give y).

Investigate further: The binomial theorem: $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$. The coefficient of $x^k y^{n-k}$ is $\binom{n}{k}$. Use this to find the coefficient of $x^2 y^3$ in $(2x-y)^5$.

4. Evaluate $\frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1}$.

Answer: 2

Method: Rationalise both fractions: $\frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} - \frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)} = (\sqrt{2}+1) - (\sqrt{2}-1) = 2$.

Investigate further: Rationalising: $\frac{1}{\sqrt{n}-1} = \frac{\sqrt{n}+1}{n-1}$ and $\frac{1}{\sqrt{n}+1} = \frac{\sqrt{n}-1}{n-1}$. Their difference is $\frac{2}{n-1}$. Verify at $n=2$: $\frac{2}{1} = 2$. ✓ Now evaluate $\frac{1}{\sqrt{5}-2} - \frac{1}{\sqrt{5}+2}$.

5. If $x = 2$ is a repeated root of $f(x) = x^3 + ax + b$, find a and b .

Answer: $a = -12$, $b = 16$

Method: $(x - 2)^2$ divides $f(x)$, so $f(x) = (x - 2)^2(x - r)$ for some r . Expanding: $x^3 - (r + 4)x^2 + (4r + 4)x - 4r$. Matching $f(x) = x^3 + ax + b$ (no x^2 term): $r + 4 = 0 \Rightarrow r = -4$. Then $a = 4(-4) + 4 = -12$ and $b = -4(-4) = 16$.

Investigate further: Alternatively: if $x = 2$ is a repeated root, then $f(2) = 0$ AND $f'(2) = 0$. $f'(x) = 3x^2 + a$. $f'(2) = 12 + a = 0 \Rightarrow a = -12$. Then $f(2) = 8 - 24 + b = 0 \Rightarrow b = 16$. This derivative approach is the standard method when calculus is allowed.

6. Simplify: $\left(\frac{2x}{y}\right)^3 \div \frac{4x^2}{y^2}$.

Answer: $\frac{2x}{y}$

Method: $\frac{8x^3}{y^3} \div \frac{4x^2}{y^2} = \frac{8x^3}{y^3} \cdot \frac{y^2}{4x^2} = \frac{2x}{y}$.

Investigate further: This simplification reveals $\left(\frac{2x}{y}\right)^3 \div \left(\frac{2x}{y}\right)^2 = \frac{2x}{y}$. The pattern: $t^3 \div t^2 = t$ for any $t = \frac{2x}{y}$. Recognising the hidden variable t saves the expansion step entirely.

7. Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.

Answer: 30

Method: $\alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta) = 6 \times 5 = 30$.

Investigate further: Factor before substituting. The expression $\alpha^2\beta + \alpha\beta^2$ has an obvious common factor $\alpha\beta$ that you should see immediately. Now find $\alpha^3\beta^2 + \alpha^2\beta^3 = \alpha^2\beta^2(\alpha + \beta)$.

8. Factorise: $4a^2 - 4ab + b^2$.

Answer: $(2a - b)^2$

Method: Perfect square trinomial with $(\alpha - \beta)^2$ form: $\alpha = 2a$, $\beta = b$.

Investigate further: Perfect square recognition: $px^2 + qxy + ry^2$ is a perfect square iff $q^2 = 4pr$. Check: $(-4)^2 = 16 = 4 \times 4 \times 1$. ✓ Identify all perfect square trinomials of the form $4a^2 + kab + 9b^2$.

9. Simplify $2^{10} - 1$ as a product of three factors greater than 1.

Answer: $3 \times 11 \times 31$

Method: $2^{10} - 1 = 1023 = (2^5 - 1)(2^5 + 1) = 31 \times 33 = 31 \times 3 \times 11$.

Investigate further: $2^{10} - 1 = (2^2 - 1)(2^8 + 2^6 + 2^4 + 2^2 + 1) = 3 \times 341 = 3 \times 11 \times 31$. Alternatively use the factorisation chain $2^{10} - 1 = (2^5 - 1)(2^5 + 1)$. Which approach is faster? Use Mersenne numbers $M_p = 2^p - 1$: are they always prime for prime p ? (No: $2^{11} - 1 = 2047 = 23 \times 89$.)

10. Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

Answer: 0

Method: $f(0) = 0^2 - 3(0) + 2 = 2$. Then $f(f(0)) = f(2) = 2^2 - 3(2) + 2 = 0$.

Investigate further: $f(f(0)) = 0$ means 0 is a period-2 point of f : $f(0) = 2$ and $f(2) = 0$, so $f(f(0)) = 0$. Explore: is 0 a fixed point? ($f(0) = 2 \neq 0$, so no.) Find all fixed points of f (where $f(x) = x$: $x^2 - 4x + 2 = 0$, $x = 2 \pm \sqrt{2}$). What does the graph of $y = f(f(x))$ look like?

Section B Manipulation Drills

1. Find A, B, C such that $x^2 + 3x + 5 \equiv A(x+1)^2 + B(x+1) + C$.

Answer: $A = 1, B = 1, C = 3$

Method: Expand the right: $A(x^2 + 2x + 1) + B(x+1) + C = Ax^2 + (2A+B)x + (A+B+C)$. Compare: $A = 1, 2A+B = 3 \Rightarrow B = 1, A+B+C = 5 \Rightarrow C = 3$. Alternatively substitute $x = -1$: $C = 1 - 3 + 5 = 3$.

Investigate further: This is expressing $x^2 + 3x + 5$ in the *shifted basis* $\{(x+1)^2, (x+1), 1\}$ centred at $x = -1$. It is equivalent to the Taylor expansion about $x = -1$: $f(-1) = 3, f'(-1) = 1, f''(-1)/2! = 1$. Compare the coefficients with $A = f''(-1)/2, B = f'(-1), C = f(-1)$.

2. The cubic $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0, p(-1) = 4, p(2) = 15$. Find a, b, c .

Answer: $a = \frac{11}{3}, b = -3, c = -\frac{5}{3}$

Method: From $p(1) = 0$: $a + b + c = -1$. From $p(-1) = 4$: $a - b + c = 5$. Subtracting gives $b = -3$, then $a + c = 2$. Now $p(2) = 15$ gives $8 + 4a + 2b + c = 15 \Rightarrow 4a + c = 13$. Subtract $a + c = 2$: $3a = 11$, so $a = \frac{11}{3}$ and $c = -\frac{5}{3}$. So the coefficients are rational, not integers.

Investigate further: Comparing coefficients (equating the polynomial at test points) is effectively solving a linear system $V\mathbf{c} = \mathbf{p}$ where V is a *Vandermonde matrix*. The Vandermonde determinant $\prod_{i>j}(x_i - x_j)$ is nonzero as long as the evaluation points are distinct, guaranteeing a unique solution.

3. Simplify: $\frac{x^3 - 8}{x^2 + 2x + 4}$.

Answer: $x - 2$

Method: Difference of cubes: $x^3 - 8 = (x-2)(x^2 + 2x + 4)$. Cancel $(x^2 + 2x + 4)$.

Investigate further: The factor $x^2 + 2x + 4$ has discriminant $4 - 16 = -12 < 0$, so it is irreducible over $\mathbb{R}$. This means $x = 2$ is the only real root of $x^3 = 8$. Generalise: $\frac{x^3 - a^3}{x^2 + ax + a^2} = x - a$ for any a .

4. State the identity $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$ and verify at $a = b = c = 1$.

Answer: At $a = b = c = 1$: LHS = $3 - 3 = 0$; RHS = $3(3 - 3) = 0$. ✓

Method: The factor $(a+b+c)$ on the RHS makes it obvious that if $a+b+c = 0$, the expression vanishes, so $a^3 + b^3 + c^3 = 3abc$. The identity is proved by expanding the RHS.

Investigate further: The second factor $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$.

So the identity shows $a^3 + b^3 + c^3 - 3abc \geq 0$ when $a + b + c \geq 0$. Can you conclude $a^3 + b^3 + c^3 \geq 3abc$ for non-negative a, b, c ? (Yes, since then $a + b + c \geq 0$.)

5. Given $a + b + c = 0$, simplify $\frac{a^3 + b^3 + c^3}{abc}$.

Answer: 3

Method: Since $a + b + c = 0$, by the identity above: $a^3 + b^3 + c^3 = 3abc$. So the ratio is 3.

Investigate further: This ratio being a constant (3) regardless of the specific values of a, b, c (subject to $a + b + c = 0$) is a non-obvious result. Try $a = 1, b = 2, c = -3$: $1 + 8 - 27 = 3(1)(2)(-3) = -18$. $\frac{-18}{-6} = 3$. ✓ Now compute $\frac{a^4 + b^4 + c^4}{(a^2 + b^2 + c^2)^2}$ when $a + b + c = 0$. (This was a WS#8 question; the answer is $\frac{1}{2}$.)

6. Verify $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n = 1, 2, 3$; hence find $\sum_{k=1}^{10} (2k - 1)^2$.

Answer: $\sum_{k=1}^{10} (2k - 1)^2 = 1330$

Method: $\sum (2k - 1)^2 = \sum (4k^2 - 4k + 1) = 4 \cdot \frac{10 \cdot 11 \cdot 21}{6} - 4 \cdot \frac{10 \cdot 11}{2} + 10 = 1540 - 220 + 10 = 1330$.

Investigate further: The sum of squares of the first n odd numbers is $\frac{n(2n-1)(2n+1)}{3}$. Verify for $n = 3$: $1 + 9 + 25 = 35 = \frac{3 \cdot 5 \cdot 7}{3} = 35$. ✓ Derive this formula from $\sum_{k=1}^n (2k - 1)^2 = 4 \sum k^2 - 4 \sum k + n$.

7. Simplify: $\frac{\frac{1}{a+b} - \frac{1}{a-b}}{\frac{1}{a+b} + \frac{1}{a-b}}$.

Answer: $-\frac{b}{a}$

Method: Numerator: $\frac{(a-b)-(a+b)}{(a+b)(a-b)} = \frac{-2b}{a^2 - b^2}$. Denominator: $\frac{2a}{a^2 - b^2}$. Ratio: $\frac{-2b}{2a} = -\frac{b}{a}$.

Investigate further: This is a *compound fraction* — a fraction whose numerator or denominator is itself a fraction. The strategy is always: simplify numerator and denominator separately before dividing. Now simplify $\frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ (this is a difference quotient for $f(x) = \frac{1}{x}$).

8. Make b the subject of $\frac{a-b}{a+b} = \frac{c-d}{c+d}$.

Answer: $b = \frac{ad}{c}$

Method: Cross-multiply: $(a-b)(c+d) = (c-d)(a+b)$. Expand: $ac + ad - bc - bd = ac + bc - ad - bd$. Simplify: $2ad = 2bc \Rightarrow ad = bc \Rightarrow b = \frac{ad}{c}$.

Investigate further: The condition $\frac{a-b}{a+b} = \frac{c-d}{c+d}$ is equivalent to $\frac{a}{b} = \frac{c}{d}$, i.e. the ratios $a : b$ and $c : d$ are equal. This is the *componendo-dividendo* rule: $\frac{p}{q} = \frac{r}{s} \Leftrightarrow \frac{p+q}{p-q} = \frac{r+s}{r-s}$. Prove it and use it to solve $\frac{x+3}{x-3} = \frac{7}{3}$ instantly.

9. Given α, β, γ are roots of $x^3 - 7x + 6 = 0$, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$.

Answer: $\alpha^2 + \beta^2 + \gamma^2 = 14$; $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = 49$

Method: Vieta: $\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = -7$, $\alpha\beta\gamma = -6$. $\alpha^2 + \beta^2 + \gamma^2 = (\sum \alpha)^2 - 2\sum \alpha\beta = 0 - 2(-7) = 14$. $(\alpha\beta + \beta\gamma + \gamma\alpha)^2 = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma) = \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 0$. So $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (-7)^2 = 49$.

Investigate further: Notice that the sum of the roots is zero ($p = 0$ for the x^2 coefficient). This means the sum of squares is simply $-2q$. Use this shortcut to instantly find the sum of squares of the roots of $x^3 - 15x + 4 = 0$.

10. Solve: $\frac{2}{x^2 - 1} - \frac{1}{x - 1} = \frac{3}{x + 1}$.

Answer: No solution (the only candidate $x = 1$ is excluded from the domain).

Method: Multiply by $(x - 1)(x + 1)$: $2(x + 1) - (x^2 - 1)/(x - 1) = 3(x - 1)$? No, $(x^2 - 1)$ is the common denominator. So $2 - (x + 1) = 3(x - 1) \Rightarrow 1 - x = 3x - 3 \Rightarrow 4x = 4 \Rightarrow x = 1$. But $x = 1$ makes the original denominators zero, so it is extraneous.

Investigate further: This is the classic case where clearing denominators produces an extraneous solution. The lesson: *always check all solutions in the original equation*. The expression $\frac{2}{x^2 - 1} - \frac{1}{x - 1}$ can be simplified to $\frac{-1}{x + 1}$ for $x \neq 1$, making the equation $\frac{-1}{x + 1} = \frac{3}{x + 1}$, which has no solution. Spot this simplification first to avoid the extraneous solution trap.

Section C Substitution & Structure

1. Solve: $\sqrt{x} - \frac{6}{\sqrt{x}} = 1$.

Answer: $x = 9$

Method: Let $t = \sqrt{x} > 0$: $t - \frac{6}{t} = 1 \Rightarrow t^2 - t - 6 = 0 \Rightarrow (t - 3)(t + 2) = 0$. Since $t > 0$: $t = 3$, so $x = 9$.

Investigate further: The substitution $t = \sqrt{x}$ converts $f(\sqrt{x}) = 0$ to a polynomial equation in t . It must be combined with the condition $t \geq 0$ to discard negative roots. Try $\sqrt{x} + \frac{2}{\sqrt{x}} = 3$ and $\sqrt{x} - \frac{1}{\sqrt{x}} = \frac{3}{\sqrt{2}}$.

2. Find the minimum of $x^2 - 6x + y^2 + 4y + 20$.

Answer: Minimum is 7, at $(x, y) = (3, -2)$.

Method: $(x^2 - 6x + 9) + (y^2 + 4y + 4) + 20 - 9 - 4 = (x - 3)^2 + (y + 2)^2 + 7 \geq 7$. Equality at $x = 3$, $y = -2$.

Investigate further: Geometrically, $(x - 3)^2 + (y + 2)^2 = r^2$ is a circle of radius r centred at $(3, -2)$. The minimum of $(x - 3)^2 + (y + 2)^2$ is 0 at the centre. So the minimum of the expression is 7 (achieved at the centre). Now: what is the minimum of $x^2 - 6x + y^2 + 4y + z^2 - 2z + 20$?

3. Show $a^4 + b^4 \geq \frac{1}{2}(a + b)^4$ is false; prove the correct inequality $a^4 + b^4 \geq \frac{1}{2}(a^2 + b^2)^2$.

Answer: Counterexample: $a = b = 1$: $2 \geq \frac{1}{2}(16) = 8$? False. Correct inequality: $(a^2 - b^2)^2 \geq 0 \Rightarrow a^4 + b^4 \geq 2a^2b^2 \Rightarrow 2(a^4 + b^4) \geq (a^2 + b^2)^2$.

Method: $\frac{1}{2}(a+b)^4 = \frac{1}{2}(a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4)$. At $a = b = 1$: $\frac{1}{2}(16) = 8 > 2 = a^4 + b^4$. So the claimed inequality is false. Correct: $(a^2 - b^2)^2 \geq 0$ gives $a^4 - 2a^2b^2 + b^4 \geq 0$, so $a^4 + b^4 \geq 2a^2b^2$, hence $2(a^4 + b^4) \geq a^4 + 2a^2b^2 + b^4 = (a^2 + b^2)^2$.

Investigate further: The correct inequality $2(a^4 + b^4) \geq (a^2 + b^2)^2$ is a special case of the *power mean inequality*: $M_4 \geq M_2$, i.e. $\left(\frac{a^4+b^4}{2}\right)^{1/4} \geq \left(\frac{a^2+b^2}{2}\right)^{1/2}$. Verify this for $a = 3, b = 1$, then for $a = 1, b = 0$.

4. Solve $x^4 - 2x^3 - 3x^2 + 4x + 4 = 0$ given that it has a repeated real root. **Method:** Try $x = 2$: $16 - 16 - 12 + 8 + 4 = 0$. ✓ So $(x - 2)^2$ divides the polynomial. Divide: $x^4 - 2x^3 - 3x^2 + 4x + 4 = (x - 2)^2(x^2 + 2x + 1)$. So $(x - 2)^2(x + 1)^2 = 0$: $x = 2$ or $x = -1$ (each a double root).

Answer: $x = 2$ (double root) and $x = -1$ (double root)

Investigate further: A polynomial $p(x)$ has a repeated root r iff both $p(r) = 0$ and $p'(r) = 0$ (the root is shared with the derivative). Check: $p'(x) = 4x^3 - 6x^2 - 6x + 4$. $p'(2) = 32 - 24 - 12 + 4 = 0$ ✓. $p'(-1) = -4 - 6 + 6 + 4 = 0$ ✓. This derivative test is the standard method for finding repeated roots.

5. Given $a, b, c > 0$ with $a + b + c = 1$, prove $ab + bc + ca \leq \frac{1}{3}$.

Answer: Proved below.

Method: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 1$. Also $a^2 + b^2 + c^2 \geq \frac{(a+b+c)^2}{3} = \frac{1}{3}$ (by C-S or AM inequality). So $2(ab + bc + ca) = 1 - (a^2 + b^2 + c^2) \leq 1 - \frac{1}{3} = \frac{2}{3}$, giving $ab + bc + ca \leq \frac{1}{3}$. □ Equality iff $a = b = c = \frac{1}{3}$.

Investigate further: Combined with the earlier result $ab + bc + ca \geq 0$, we have $0 \leq ab + bc + ca \leq \frac{1}{3}$ whenever $a, b, c > 0$, $a + b + c = 1$. The lower bound 0 is approached (but not achieved for positive reals) as one variable dominates. The upper bound $\frac{1}{3}$ is achieved at $a = b = c$.

6. If $x + y = 1$, find the maximum of $x^2y + xy^2$.

Answer: $\frac{1}{4}$

Method: $x^2y + xy^2 = xy(x + y) = xy$. By AM-GM: $xy \leq \left(\frac{x+y}{2}\right)^2 = \frac{1}{4}$. Equality at $x = y = \frac{1}{2}$.

Investigate further: The expression $x^2y + xy^2 = xy(x + y)$ factors instantly — this is the key recognition. Without seeing the factor, one might substitute $y = 1 - x$ and differentiate (using calculus), but the algebraic route is much faster. Always factor symmetric-looking expressions before substituting.

7. Solve over $\mathbb{R}$: $\sqrt{3x+1} - \sqrt{x+4} = 1$.

Answer: $x = 5$ only

Method: Isolate $\sqrt{3x+1} = 1 + \sqrt{x+4}$ and square: $3x+1 = x+5+2\sqrt{x+4} \Rightarrow x-2 = \sqrt{x+4}$. Hence $x \geq 2$. Square again: $(x-2)^2 = x+4 \Rightarrow x^2 - 5x = 0 \Rightarrow x = 0$ or $x = 5$. Domain $x \geq 2$ leaves $x = 5$. Check: $\sqrt{16} - \sqrt{9} = 1$.

Investigate further: The squaring step introduced $x = 0$ as an extraneous solution. When we required $x - 2 = \sqrt{x+4} \geq 0$, i.e. $x \geq 2$, this eliminated $x = 0$ before even checking. Spotting the

domain restriction *before* squaring saves a check step. Practise identifying domain restrictions at each squaring step.

8. For what values of b, c do $x^2 + bx + c = 0$ and $x^2 + cx + b = 0$ have the same roots?

Answer: $b = c$

Method: For both quadratics to have the exact same roots, all corresponding coefficients must be equal (since both are monic). Therefore $b = c$.

Investigate further: The two equations have the *same* roots only when $b = c$. But they could each have *one common root*: if r is a shared root, then $r^2 + br + c = 0$ and $r^2 + cr + b = 0$. Subtracting: $(b - c)r + (c - b) = 0 \Rightarrow (b - c)(r - 1) = 0$. So either $b = c$ or $r = 1$. If $r = 1$: $1 + b + c = 0$, so $b + c = -1$.

Section D Challenge

(SMC / BMO1 difficulty)

1. **Schur's Inequality (degree 2):** Prove $\sum a^2(a - b)(a - c) \geq 0$ for $a, b, c \geq 0$.

Answer: Proved below.

Method: WLOG $a \geq b \geq c \geq 0$. Group as $(a - b)[a^2(a - c) - b^2(b - c)] + c^2(c - a)(c - b)$. Since $c \geq 0$, $(c - a) \leq 0$ and $(c - b) \leq 0$, so the second term is ≥ 0 . For the first term, since $a \geq b \geq c \geq 0$, we have $a - c \geq b - c \geq 0$ and $a^2 \geq b^2$, so $a^2(a - c) \geq b^2(b - c)$. Thus the bracket is ≥ 0 , and $(a - b) \geq 0$. The sum is ≥ 0 .

Investigate further: Schur's inequality for general $t \geq 0$ states $a^t(a - b)(a - c) + b^t(b - a)(b - c) + c^t(c - a)(c - b) \geq 0$. The $t = 1$ case gives: $a^3 + b^3 + c^3 + 3abc \geq (ab(a + b) + bc(b + c) + ca(c + a))$. This is a BMO1-level inequality. Look up the proof using $t = 1$ and try to verify it numerically for $a = 3, b = 2, c = 1$.

2. Find all polynomials with real coefficients satisfying $p(x^2) = p(x)^2$.

Answer: $p(x) = 0$ and $p(x) = x^n$ for non-negative integers n .

Method: Suppose p has degree d . Then $p(x^2)$ has degree $2d$ and $p(x)^2$ has degree $2d$: consistent. If r is a root of p , then $p(r^2) = p(r)^2 = 0$, so r^2 is also a root. The sequence $r, r^2, r^4, \dots$ must terminate (finitely many roots), so $r = 0$ or $r = 1$ or $r = -1$. If $r = -1$: $r^2 = 1$ is a root, and $1^2 = 1$, so $p(1) = p(1)^2 \Rightarrow p(1) = 0$ or 1 . Careful analysis gives only monomials $p(x) = x^n$ and the zero polynomial.

Investigate further: Functional equations for polynomials are a rich topic. Try $p(x + 1) - p(x) = nx^{n-1}$ (the discrete derivative). The polynomial that satisfies this has leading term x^n (related to Bernoulli polynomials). Also explore: find all polynomials satisfying $p(x^2 - 1) = p(x)p(-x)$.

3. Find a closed form for $\frac{1^2}{1 \cdot 3} + \frac{2^2}{3 \cdot 5} + \dots + \frac{n^2}{(2n - 1)(2n + 1)}$.

Answer: $\frac{n(n + 1)}{2(2n + 1)}$

Method: $\frac{k^2}{(2k - 1)(2k + 1)} = \frac{1}{4} \cdot \frac{4k^2}{(2k - 1)(2k + 1)} = \frac{1}{4} \cdot \frac{(2k - 1)(2k + 1) + 1}{(2k - 1)(2k + 1)} = \frac{1}{4} \left(1 + \frac{1}{(2k - 1)(2k + 1)} \right)$. Sum $= \frac{n}{4} + \frac{1}{4} \sum_{k=1}^n \frac{1}{(2k - 1)(2k + 1)} = \frac{n}{4} + \frac{1}{4} \cdot \frac{n}{2n + 1} = \frac{n}{4} \cdot \frac{2n + 2}{2n + 1} = \frac{n(n + 1)}{2(2n + 1)}$.

Investigate further: Verify for $n = 1$: $\frac{1}{3} = \frac{1 \cdot 2}{2 \cdot 3} = \frac{1}{3}$. ✓ For $n = 2$: $\frac{1}{3} + \frac{4}{15} = \frac{5+4}{15} = \frac{9}{15} = \frac{3}{5} = \frac{2 \cdot 3}{2 \cdot 5}$.
 ✓ Now find $\sum_{k=1}^n \frac{k^2}{(k-1)(k+1)}$ for $k \geq 2$ using the same decomposition technique.

4. A polynomial $p(x)$ of degree 3 satisfies $p(n) = \frac{1}{n}$ for $n = 1, 2, 3, 4$. Find $p(5)$.

Answer: $p(5) = 0$

Method: Consider $q(x) = xp(x) - 1$. Then $q(n) = n \cdot \frac{1}{n} - 1 = 0$ for $n = 1, 2, 3, 4$, so q is a quartic with roots 1, 2, 3, 4: $q(x) = c(x-1)(x-2)(x-3)(x-4)$. At $x = 0$: $q(0) = -1 = 24c$, so $c = -\frac{1}{24}$. Hence $q(5) = -\frac{1}{24}(4)(3)(2)(1) = -1$. But $q(5) = 5p(5) - 1$, so $5p(5) - 1 = -1$, giving $p(5) = 0$.

Investigate further: So $p(5) = 0$ means $x = 5$ is a root of p . This is a striking result! Explore: compute $p(0)$. From $q(0) = 0 \cdot p(0) - 1 = -1$ and $q(0) = c \cdot (-1)(-2)(-3)(-4) = 24c = -1$, so $c = -\frac{1}{24}$. Then $p(x) = \frac{q(x)+1}{x} = \frac{-\frac{1}{24}(x-1)(x-2)(x-3)(x-4)+1}{x}$ for $x \neq 0$.

5. Find all pairs of positive integers (m, n) with $m > n$ such that $m^2 - n^2 = 105$.

Answer: $(m, n) \in \{(53, 52), (19, 16), (13, 8), (11, 4)\}$

Method: $(m-n)(m+n) = 105 = 3 \times 5 \times 7$. Set $m-n = a$, $m+n = b$: $ab = 105$, $b > a > 0$, same parity (both odd since 105 is odd and a, b have same parity). Factor pairs of 105 with $a < b$, both odd: $(1, 105), (3, 35), (5, 21), (7, 15)$. Each gives $m = \frac{a+b}{2}$, $n = \frac{b-a}{2}$: $(1, 105) \rightarrow (53, 52)$; $(3, 35) \rightarrow (19, 16)$; $(5, 21) \rightarrow (13, 8)$; $(7, 15) \rightarrow (11, 4)$.

Investigate further: The key observation: since 105 is odd, $m-n$ and $m+n$ must both be odd (both even would give $4 \mid (m^2 - n^2)$, but $4 \nmid 105$). This parity constraint halves the number of cases. Generalise: for which N does $m^2 - n^2 = N$ have no positive integer solutions? (When $N \equiv 2 \pmod{4}$ or $N = 1$ or $N = 4$.)

Top 5 Patterns — Worksheet #6

- Shifting basis: $p(x)$ in powers of $(x-a)$.** Evaluate $p(a)$, $p'(a)$, $p''(a)$ etc. to find coefficients without expanding. Equivalent to Taylor expansion at $x = a$.
- Substitution $t = \sqrt{x}$ for surd equations.** Always combine with the constraint $t \geq 0$ to discard extraneous roots before they enter the algebra.
- The identity $a^3 + b^3 + c^3 - 3abc = (a+b+c)(\dots)$.** The zero of the first factor ($a+b+c=0$) is the most useful case. Memorise both the identity and the special case.
- Polynomial masking: $q(x) = xp(x) - 1$ for rational interpolation.** When $p(n) = 1/n$, multiply through to clear denominators, find the masked polynomial's roots, determine its leading coefficient from $q(0) = -1$.
- Factoring $m^2 - n^2 = (m-n)(m+n)$ and using parity.** The parity of $m \pm n$ is determined by the parity of $m^2 - n^2$. Use this to eliminate half the factor pairs immediately.

Common Traps — Worksheet #6

- Extraneous solutions after clearing denominators.** In B10, $x = 1$ satisfies the cleared equation but not the original. Always substitute back.
- Not checking both cases when squaring radical equations.** In C7, squaring $x-2 = \sqrt{x+4}$

requires $x \geq 2$: this eliminates $x = 0$ before checking.

3. **Confusing “same roots” with “same polynomial.”** In C8, two quadratics can share all roots only if all coefficients are equal (for monic quadratics). Carefully set up the Vieta conditions for both.
4. **Applying $a^4 + b^4 \geq \frac{1}{2}(a + b)^4$.** This is false. The correct form is $2(a^4 + b^4) \geq (a^2 + b^2)^2$, which is weaker. Test inequalities with $a = b = 1$ before claiming them.
5. **Forgetting that $p(x^2) = p(x)^2$ constrains roots severely.** The functional equation forces the root set to be closed under squaring, leaving only $\{0\}$, $\{1\}$, $\{0, 1\}$, or empty. Don’t guess a general polynomial form.

“In mathematics the art of proposing a question must be held of higher value than solving it.” — Georg Cantor

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths